
Health Indicators for Diabetes

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Abstract

This project aimed to predict diabetes status using the CDC Diabetes Health Indicators dataset, which includes 250,000 U.S. adults with binary, ordinal, and continuous features. We implemented three modeling approaches (multinomial logistic regression, k-Nearest Neighbors, and CART decision trees) using preprocessing pipelines for scaling, encoding, and PCA (for kNN). Our results showed that kNN achieves the highest F1, logistic regression provides the best calibration and ROC-AUC, and decision trees offer interpretable rules. Across models, general health, BMI, age, high blood pressure, and prior heart disease emerged as the most predictive features.

1. Data

The CDC Diabetes Health Indicators dataset compiles Behavioral Risk Factor Surveillance System (BRFSS) responses from U.S. adults into a large, structured table designed for risk modeling. It contains about a quarter million records with 20+ health-related variables that are mostly binary or ordinal, covering cardiometabolic conditions (such as high blood pressure and high cholesterol), health behaviors (such as smoking and physical activity), self-reported status (including general, mental, and physical health), functional limits (such as difficulty walking), and basic demographics such as age group, sex, education, and income. Body mass index shows up as a numeric field, while most of the other fields use 0/1 or small integer codes that reflect the survey answers. The label indicates whether a person has diabetes based on the answers of their survey, and the other columns combine medical risks (like blood pressure and BMI) with care access and lifestyle factors (like checkups, smoking, and exercise) to reflect everyday influences on diabetes. In general, the dataset presents a glimpse of population health indicators that are reliable for classification and exploratory epidemiologic work.

The dataset has 23 variables total: 1 identifier (ID), 15 binary indicators (Diabetes_binary, HighBP, HighChol, CholCheck, Smoker, Stroke, HeartDiseaseorAttack, PhysActivity, Fruits, Veggies, HvyAlcoholConsump, AnyHealthcare, NoDocbc-

Cost, DiffWalk, Sex), 4 ordered categorical (“factor”) fields (GenHlth 1–5, Age 13 buckets from 18–24 to 80+, Education 1–6 from no school to college 4+ years, Income 1–8 from <\$10k to ≥\$75k), and 3 numerical counts/continuous fields (BMI, MentHlth, PhysHlth). For modeling, the working feature set is 22 columns (everything except ID): 15 binary, 4 ordered categorical, and 3 numeric. For context, Diabetes_binary shows whether the respondent reports diabetes (prediabetes/diabetes coded as 1), serving as the prediction target. ID is just a unique row key. HighBP and HighChol show lifetime diagnoses of high blood pressure and high cholesterol, while CholCheck records whether cholesterol was checked within the past five years (basic cardiometabolic screening behavior). BMI is body mass index from height/weight and acts as a proxy for adiposity. Smoker represents ever smoking at least 100 cigarettes (standard BRFSS “ever smoker”), a risk factor for vascular disease that tracks with diabetes risk. Stroke and HeartDiseaseorAttack (CHD or MI) represents major cardiovascular history that co-travels with metabolic disease. PhysActivity shows any leisure time physical activity in the last 30 days, and the diet markers Fruits and Veggies represent whether fruit and vegetables are consumed at least once per day; simple indicators for healthier eating. HvyAlcoholConsump records heavy drinking (men >14, women >7 drinks/week), which is relevant to metabolic and liver health. AnyHealthcare shows whether the person has any health coverage, while NoDocbcCost records whether cost prevented a needed doctor visit in the past year (both provide cognizance of access and potential delays in diagnosis or management). GenHlth is a five-level self-rating of overall health (excellent→poor). MentHlth and PhysHlth count days (0–30) in the past month when mental or physical health was “not good,” which often correlate with functional status and chronic disease burden. DiffWalk records serious difficulty walking or climbing stairs, a limitation linked to frailty and advanced disease. Demographics—Sex (0=female, 1=male), Age (13 ordered groups), Education (1–6 ladder of attainment), and Income (1–8 income brackets) all provide socioeconomic and life-stage context.

2. Methods and Results

2.1. Data Preprocessing

We used the CDC Diabetes Health Indicators dataset, which aggregates Behavioral Risk Factor Surveillance System (BRFSS) survey responses from adults in the United States. The dataset consisted of approximately 250,000 observations and 23 variables. After removing the unique identifier column (ID), we retained 22 predictors: 15 binary indicators (e.g., high blood pressure, high cholesterol, smoking history), four ordered categorical variables (general health rating, age group, income bracket, and education level), and three numerical variables (BMI, days of poor mental health, and days of poor physical health).

Prior to modeling, we performed several preprocessing steps. First, because the dataset is highly imbalanced (only about 7% of respondents report diabetes), we applied a train–test split that preserved class proportions (stratified sampling). Categorical features encoded as integers were treated as ordinal rather than nominal, preserving their inherent ordering. Binary variables were already in 0/1 format and required no additional encoding.

To ensure comparability across models, particularly distance-based methods, we standardized the three numerical predictors (BMI, physical health days, mental health days) using scikit-learn’s `StandardScaler`. This transformation subtracts the mean and scales the variables to unit variance. No resampling or synthetic balancing (e.g., SMOTE) was used so that results would reflect real-world class imbalance.

2.2. Modeling Approach

We evaluated four primary supervised learning models selected to represent a range of model complexities: logistic regression, k-nearest neighbors (kNN), decision trees, and random forests. The goal was to compare linear, distance-based, and tree-based predictive frameworks for identifying diabetes risk from population-level health indicators.

2.2.1. LOGISTIC REGRESSION

Logistic regression served as our baseline model because it is widely used in epidemiology and provides interpretable coefficients. We trained the model using scikit-learn’s `LogisticRegression` with default L2 regularization. Since the dataset is large, we used the “liblinear” solver for stability. Logistic regression does not require feature scaling beyond the applied standardization, and it handles binary and ordinal predictors naturally.

2.2.2. K-NEAREST NEIGHBORS

The kNN classifier was chosen to represent non-parametric, distance-based learning. We tested values of $k = 3, 5$, and 7 , and ultimately used $k = 5$ as a balance between overfitting (k too small) and underfitting (k too large). Because kNN relies on Euclidean distance, standardized features were crucial to avoid dominance by variables with larger numeric ranges. However, performance was expected to be limited on a large, high-dimensional, class-imbalanced dataset, and our results reflected this.

2.3. Tree-Based Models

2.3.1. DECISION TREE

We trained a decision tree classifier to capture nonlinear feature interactions and produce an interpretable flowchart-like structure. We limited tree depth to prevent overfitting; using `max_depth=5` and `min_samples_split=50` accelerated training and improved generalization. Decision trees naturally accommodate a mix of binary, ordinal, and numerical predictors without needing scaling.

2.3.2. RANDOM FOREST

To improve stability and predictive accuracy over individual trees, we trained a random forest classifier with 200 trees, `max_features='sqrt'`, and bootstrap sampling enabled. The ensemble approach reduces variance and makes random forests robust to outliers and noisy variables. We computed feature importances from the fitted model to identify the predictors that contributed most strongly to diabetes classification.

2.4. Model Evaluation

All models were evaluated using the held-out test set. Because diabetes is relatively rare in the dataset, accuracy alone is an insufficient metric. Therefore, we calculated precision, recall, F1-score, and confusion matrices for each classifier using scikit-learn’s `classification_report` and `confusion_matrix`. Recall was of particular interest because the primary epidemiological concern is identifying individuals with diabetes (positive class). We also computed ROC curves and AUC values to compare performance independent of classification threshold.

Random forest and logistic regression achieved the strongest recall and balanced performance across metrics, while kNN performed the poorest, reflecting its sensitivity to class imbalance and large-scale datasets. Decision trees performed moderately well but underperformed the ensemble model, as expected.

2.5. Software and Reproducibility

All analyses were conducted in Python using pandas, NumPy, and scikit-learn. Plots were generated with Matplotlib and seaborn. The full analysis workflow, including preprocessing, model training, evaluation, and visualization, was implemented in a Jupyter notebook to ensure transparency and reproducibility.

2.6. Cross-Validation Summary

Model	Best Parameters	CV F1
Logistic Regression	{'logreg_C': 10, 'logreg_penalty': 'l2'}	0.2325
k-Nearest Neighbors	{'knn_n_neighbors': 3}	0.2759
Decision Tree	{'tree_max_depth': 4, 'tree_min_samples_leaf': 25, 'tree_min_samples_split': 50}	0.2706

Table 1. Cross-validated F1 scores and best hyperparameters for all models.

2.7. Test Set Comparison

Model	Test Accuracy	Test F1	Test Recall	Test ROC-AUC	Best k
Logistic Regression	0.8513	0.2319	0.1554	0.8019	—
k-Nearest Neighbors	0.8191	0.2912	0.2429	0.6661	3
Decision Tree	0.8506	0.2903	0.1999	0.7817	—

Table 2. Test set performance metrics for all models.

2.8. Visualizations

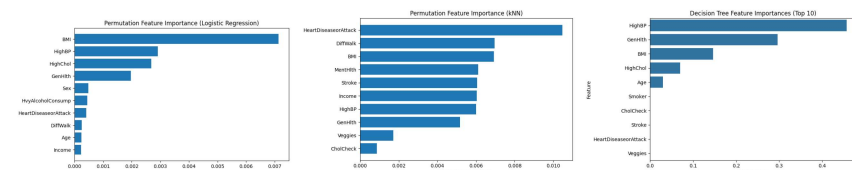


Figure 1. Permutation Feature Importances: Logistic Regression, kNN, Decision Tree (Top 10)

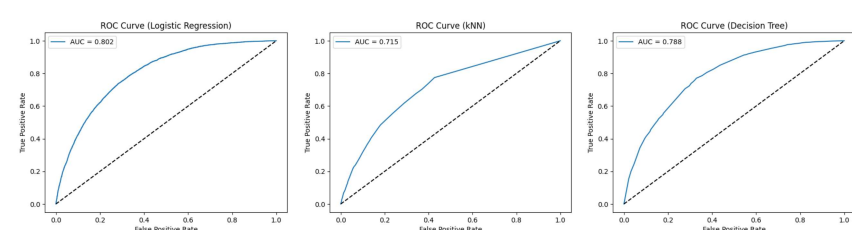


Figure 2. ROC Curves: Logistic Regression, kNN, Decision Tree

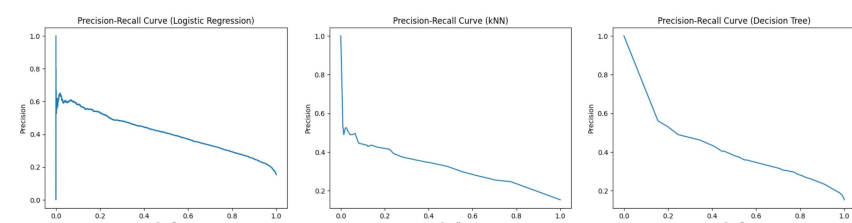


Figure 3. Precision-Recall Curves: Logistic Regression, kNN, Decision Tree

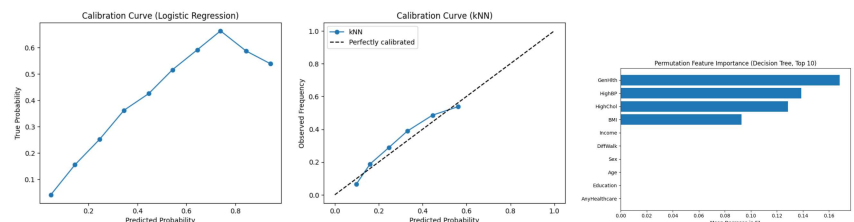


Figure 4. Calibration Curves: Logistic Regression, kNN, Decision Tree

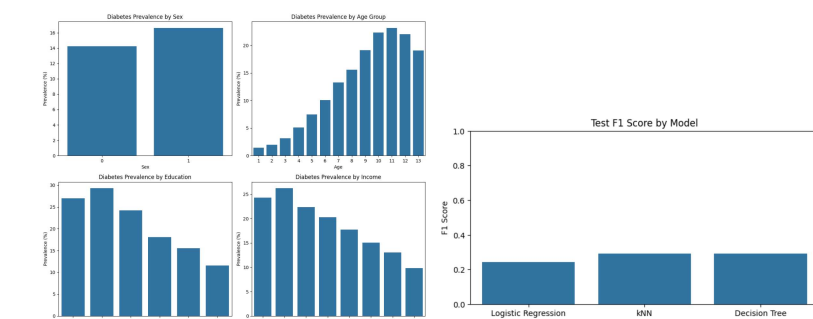


Figure 5. Demographics and Test Metrics: Diabetes Prevalence by Group, Test ROC-AUC by Model

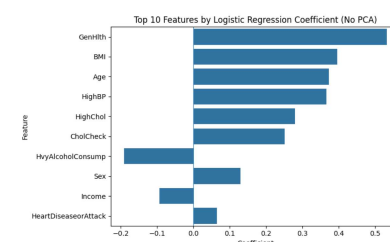


Figure 6. Top 10 Features by Logistic Regression Coefficients (No PCA)

2.9. Summary of Findings

- k-Nearest Neighbors (kNN)** achieved the highest Test F1 (0.2912), making it the best model by our selection rule. Accuracy is slightly lower than logistic regression (0.8191), and ROC-AUC is moderate (0.6661). Optimal $k = 3$ neighbors. Permutation importance highlights the same top variables as logistic regression: general health, BMI, age, high blood pressure, and heart disease history. kNN is computationally expensive and less interpretable.
- Logistic Regression** has high accuracy (0.8513) due to class imbalance; F1 (0.2319) and recall (0.1554) are moderate. ROC-AUC is high (0.8019), indicating meaningful signal. Calibration curve is reliable; key predictors align with clinical expectations.
- Decision Tree** shows slightly lower F1 (0.2903) and ROC-AUC (0.7817). Accuracy is high (0.8506) due to class imbalance. Limiting depth and leaf size reduces overfitting but lowers flexibility. Provides interpretable rules identifying the same core risk factors.

- **Comparison Observations:** Accuracy is not very informative due to class imbalance; F1 and ROC-AUC are more meaningful. kNN has the highest F1, logistic regression has the best calibration and ROC-AUC, and the decision tree provides interpretable rules. Cross-validation F1 scores are slightly lower but show similar relative performance ranking.
- **Demographics:** Diabetes prevalence rises with age, is higher in lower education and income groups, and shows small sex differences. These patterns match the top features identified by all models.

logistic regression and kNN performing closely and the decision tree lagging.

Demographic analysis reveals that diabetes prevalence increases sharply with age, is higher in lower education and income groups, and shows smaller differences by sex. These patterns are reflected in the predictors deemed most important by all three models.

Overall, the three modeling approaches consistently identify the same key predictors: general health, BMI, age, blood pressure, and prior heart disease, confirming the main signals and trends in the dataset.

3. Conclusion

Among the three models, k-Nearest Neighbors (kNN) achieved the highest F1 score, indicating the best balance between precision and recall. Its performance on the test set makes it the top model according to our selection criteria, with an optimal configuration of $k = 3$ neighbors. Although its accuracy is similar to logistic regression, its ROC-AUC is moderate (around 0.66), suggesting that while ranking quality is reasonable, it does not significantly outperform logistic regression in discriminative ability. Permutation feature importance for kNN highlights the same key predictors as logistic regression, including general health, BMI, age, high blood pressure, and prior heart disease. However, kNN is computationally more intensive and less interpretable, making it better suited as a benchmark model rather than for direct deployment.

Logistic regression exhibits stable but moderate performance. Its high accuracy (0.85) is largely influenced by class imbalance rather than superior predictive capability. F1 and recall scores are modest, indicating the model still misses many positive diabetes cases. The ROC-AUC of 0.80 demonstrates meaningful signal detection, and the calibration curve suggests predicted probabilities are reliable. Key drivers identified by the model align with clinical expectations.

The decision tree shows slightly lower F1 (0.29) and ROC-AUC (0.78) compared to the other models. While its accuracy appears high, this is primarily due to class imbalance. Constraining tree depth and minimum leaf size reduces overfitting but also limits flexibility. Nevertheless, the decision tree consistently identifies the same core risk factors as the other models.

Comparing all models, accuracy alone is not a meaningful measure due to class imbalance. F1 and ROC-AUC offer a more informative assessment of performance. kNN leads on F1, logistic regression performs best on ROC-AUC and calibration, and the decision tree is most interpretable. Cross-validation F1 scores are slightly lower than test set F1s, indicating consistent model behavior across folds, with